

Assignment No.7

Problem Statement : Cursors: (All types: Implicit, Explicit, Cursor FOR Loop, Parameterized Cursor)

Program :

```
CREATE TABLE O_RollCall(name VARCHAR(20),roll NUMBER(10),  
class VARCHAR(20));
```

```
CREATE TABLE N_RollCall(name VARCHAR(20),roll NUMBER(10),  
class VARCHAR(20));
```

Inserting data

```
INSERT INTO O_RollCall VALUES ('Stewie', 1, 'Comp 1');
```

```
INSERT INTO O_RollCall VALUES ('Edie', 2, 'Comp 2');
```

```
INSERT INTO O_RollCall VALUES ('Stomp', 3, 'Comp 3');
```

```
INSERT INTO N_RollCall VALUES ('Stewie', 1, 'Comp 1');
```

```
INSERT INTO N_RollCall VALUES ('Edie', 2, 'Comp 2');
```

```
INSERT INTO N_RollCall VALUES ('Stomp', 3, 'Comp 3');
```

```
INSERT INTO N_RollCall VALUES ('Gundeti', 6, 'Comp 3');
```

```
INSERT INTO N_RollCall VALUES ('Kalas', 7, 'Comp 2');
```

Procedure

Explicit cursor

```
DECLARE
```

```
    p_name VARCHAR(20);
```

```
    p_rollno NUMBER(10);
```

```
    p_class VARCHAR(20);
```

```
    CURSOR cc1 IS SELECT * FROM N_RollCall WHERE roll NOT IN (SELECT roll  
FROM O_RollCall);
```

```
BEGIN
```

```
    OPEN cc1;
```

```
    LOOP
```

```
        FETCH cc1 INTO p_name, p_rollno, p_class;
```

```
        INSERT INTO O_RollCall VALUES (p_name, p_rollno, p_class);
```

```
    EXIT WHEN cc1%notfound;
```

```

        DBMS_OUTPUT.PUT_LINE(p_name || ' ' || p_rollno || ' ' || p_class);
    END LOOP;
    CLOSE cc1;
END;
/

```

Statement processed. Gundeti 6 Comp 3 Kalas 7 Comp 2

O_RollCall table after procedure:

NAME	ROLL	CLASS
Stewie	1	Comp 1
Edie	2	Comp 2
Stomp	3	Comp 3
Gundeti	6	Comp 3
Kalas	7	Comp 2
Kalas	7	Comp 2

Parameterized cursor

```

DECLARE
    p_name VARCHAR(20);
    p_rollno NUMBER(10);
    p_class VARCHAR(20);
    CURSOR pp1(roll1 NUMBER) IS SELECT * FROM N_RollCall WHERE roll > roll1;
BEGIN
    OPEN pp1(3);
    LOOP
        FETCH pp1 INTO p_name, p_rollno, p_class;
        EXIT WHEN pp1%notfound;
        DBMS_OUTPUT.PUT_LINE(p_name || ' ' || p_rollno || ' ' || p_class);
    END LOOP;
    CLOSE pp1;
END;
/

```

Statement processed. Gundeti 6 Comp 3 Kalas 7 Comp 2

N_RollCall table after procedure:

NAME	ROLL	CLASS
Stewie	1	Comp 1
Edie	2	Comp 2
Stomp	3	Comp 3
Gundeti	6	Comp 3
Kalas	7	Comp 2

Implicit Cursor

DECLARE

total_rows NUMBER(2);

BEGIN

UPDATE N_RollCall SET roll = roll + 1;

IF sql%notfound THEN

dbms_output.put_line('no roll was updated');

ELSIF sql%found THEN

total_rows := sql%rowcount;

DBMS_OUTPUT.PUT_LINE(total_rows || ' roll calls were affected ');

END IF;

END;

/

Statement processed. 7 roll calls were affected

N_RollCall table after procedure:

NAME	ROLL	CLASS
Stewie	2	Comp 1
Edie	3	Comp 2
Stomp	6	Comp 3
Gundeti	7	Comp 3
Kalas	8	Comp 2